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Apr 5, 2024 • 25 tweets • cirnosad/status/1776387643951170044

Beyond the amateur war analysis: Game theory in the context of Israel's attack on Iran.

Now that we have determined what Israel did and who it killed, the next most pertinent question is why and "why now"?
The answers lie in a little discussed field.
/🧵

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War Intro Thread:
For the first time, Iran was struck by Israel directly and overtly.
It happened in a third territory: war-torn Syria, but in Iranian territory, an embassy annex.
Official military attaches were killed, very senior officers in charge of logistical operations./🧵

11:58 AM · Apr 5, 2024

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Game theory is defined as the analysis of strategy when confronting decision making that occurs in:
* Discrete steps or in continuous fashion.
* Finite or infinite duration.

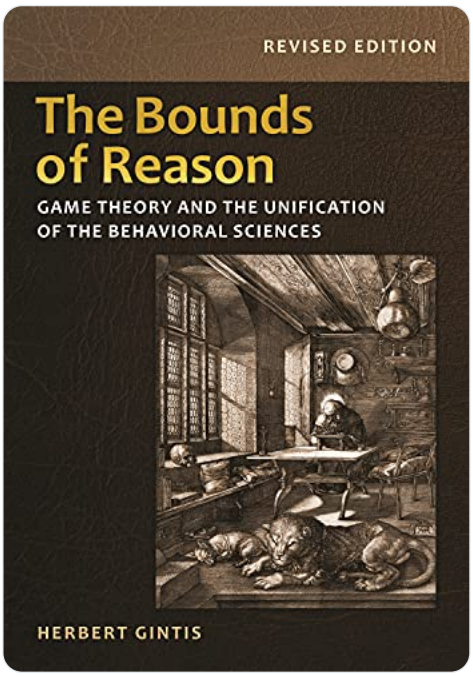
* Perfect or imperfect information.

With an understanding of the psychology & the behavior of the players.

Game theory poses as a mathematical model or field, but it is in fact a purely psychological one with some mathematical window dressing.

It has been scoffed at by many as useless in the real world, especially in the economic context it was so often employed in.

I disagree!



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Why game theory never will be anything but a footnote in the history of social science

Lars Pålsson Syll [Malmö University, Sweden]

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> "I have no fellow-feeling with those economic theorists who, off the record at seminars and conferences, admit that they are only playing a game with other theorists. If their models are not intended seriously … why do they expect me to spend my time listening to their expositions? Count me out of the game" (Robert Sugden).

Game theory is an axiomatic-mathematical theory that presents a set of axioms that people have to 'satisfy' by definition to count as 'rational.' This makes for 'rigorous' and 'precise' conclusions – but never about the real world. *Game theory* does not give us any information at all about the real world. Instead of confronting the theory with real-world phenomena it becomes a simple matter of definition if real-world phenomena are to count as signs of 'rationality.' It gives us absolutely irrefutable knowledge – but only since the knowledge is purely definitional.

> "Mathematical theorems are tautologies. They cannot be false because they do not say anything substantive. They merely spell out the implications of how things have been defined. The basic propositions of game theory have precisely the same character" (Binmore, 1994: 23).

Despite its uselessness *as a mathematical theory*, it is a very powerful psychological theory and framework and I believe it is worthy of your time and consideration.

Let's start with a single *classical* example, then demonstrate its consequences: The prison's dilemma.

There are in fact several formulations of it, I'm going to give you the normie one one first.

You & an accomplice are in jail & are up for trial.
A lawyer walks into your cell and you are given a plea bargain offer if you confess your crime. You cannot communicate w/ each other!

You work something interesting out: the lawyer lacks strong evidence and if neither of you confess, you will both be freed. If you both confess you both get the plea bargain, giving you a 3yr sentence. If only one confesses, there will be enough evidence to put you away for 20yr.

		Accomplice's choice	
		Remain Silent	Confess
Your choice	Remain Silent	Freedom for both!	You go to jail for 20yr, he gets 3yr.
	Confess	You go to jail for 3 yr, he gets 20yr.	Both of you get 3 years

Remaining silent and not accepting the plea bargain has a huge upside: you'll be freed, as long as he doesn't confess. But you don't happen to know your accomplice very well and can't guess what he would do, should he confess you'd be screwed.
Confessing seems safer: 3yr vs 20yr!

If you have no information and the possibility of your partner not confessing is 50%, you are putting 17 years at stake for a coin toss to regain 3.
John Nash would claim this is the equilibrium strategy:
Neither of you benefit from changing course, you will BOTH confess.

Such a shame right? Both of you working against your own interests. Perhaps if you knew them better you would remain silent, as you would have discussed exactly this scenario earlier.

But not all as it seems. There's something missing in this scenario, if you read it well.

You both used information you gleaned from the lawyer: That they lacked evidence. This helped you create the pay off matrix. You also lacked the information you could have used: your accomplice's preferences and behaviour. You may have misread them: they may always remain silent.

The simplistic notion of probabilities and payoffs, and a binary choice also rarely map to real life decision making. While the choice the prisoners face is important, it won't be their last. Indeed, there's a life in and after prison they will both have to deal with.

For example, let's say you find out that you're going to go to the same prison (if you go to a prison). You can imagine what will happen to one of you if you confess and the other remains silent. You're going to get shanked. The "cheap" 3 years becomes a death sentence!

Real scenarios are continuous, involve non-discrete steps, a larger decision set (e.g. you could plan a break out and not confess nor remain silent) than are formulated. Even formulating the matrix and making a conclusion from it may change your preferences and thus likelihoods!

This was very obvious to the people who formulated the original "prisoner's dilemma", which was very different from the normie one.

It was in fact formulated by the RAND corporation, by Flood and Dresher, the Flood-Dresher experiment... They set about to prove Nash wrong!

F&D recruited two people, a UCLA graduate Armen Alchian (AA) and a colleague from RAND, John D. Williams (JW). They set up a game with an unfair payoff and two decisions:
- To "cooperate"
- To "defect"
The game would be played 100 times & the AA/JW would write running commentary.

(AA's payoff, JW's payoff)	JW's Strategy 1 [Defect]	JW's Strategy 2 [Cooperate]
AA's Strategy 1 [Cooperate]	(-1, 2)	(1/2, 1)
AA's Strategy 2 [Defect]	**(0, 1/2)**	(1, -1)

It's immediately obvious that the Nash equilibrium (the strategy which neither player should change as they cannot maximise pay-off) is for both to defect. AA gets no payoff and JW gets a minimal payoff. Naive game theory would thus dictate that the players would NOT cooperate...

... but that's not what happened! Not even close.
After some scuffles, JW "teaches" AA to cooperate, through generosity and tit-for-tat moves.

In the end, mutual cooperation happened 60% of the time. This confused John Nash, who was shown the result of the experiment...

Game	*Moves*		*Armen Alchian's comments*	*John Williams's comments*
	AA	*JW*		
1	D	C	JW will play D—sure win. Hence if I play C—I lose.	Hope he's bright.
2	D	C	What is he doing ?!!	He isn't but maybe he'll wise up
3	D	D	Trying mixed ?	Okay, dope.
4	D	D	Has he settled on D ?	Okay, dope.
5	C	D	Perverse!	It isn't the best of all possible worlds.
6	D	C	I'm sticking to D since he will mix for at least 4 more times.	Oh ho! Guess I'll have to give him another chance.
7	D	C		Cagey, ain't he ? Well...
8	D	D		In time he could learn, but not in ten moves so:
9	D	D	If I mix occasionally, he will switch—but why will he ever switch from D ?	
11	D	C		Probably learned by now.
12	C	C		I'll be damned ! But I'll try again.
13	C	C		That's better.
14	C	C		Ha!
15	C	C		(bliss)
16	D	C		
17	C	D		The stinker.
18	C	D		He's crazy. I'll teach him the hard way.
19	D	D	I'm completely confused. Is he trying to convey information to me ?	Let him suffer.
20	C	D		
21	C	C		Maybe he'll be a good boy now.
22	D	C		Always takes time to learn.
23	C	C		Time.
24	C	C		
25	C	C		
26	D	C		
27	C	D		Same old story.

John Nash... quickly grabbed onto the last move: They both defected.

He pointed out his theory is thus still valid, the last move means there's no retaliation and a build up of information afterwards. So both players will take the most "rational" choice and defect. Sounds right?

91	C	C	When will he switch as a last minute grab of D ? Can I beat him to it as late as possible ?	
92	C	C		Good.
93	C	C		
94	C	C		
95	C	C		
96	C	C		
97	C	C		
98	C	C		
99	D	C		
100	D	D		

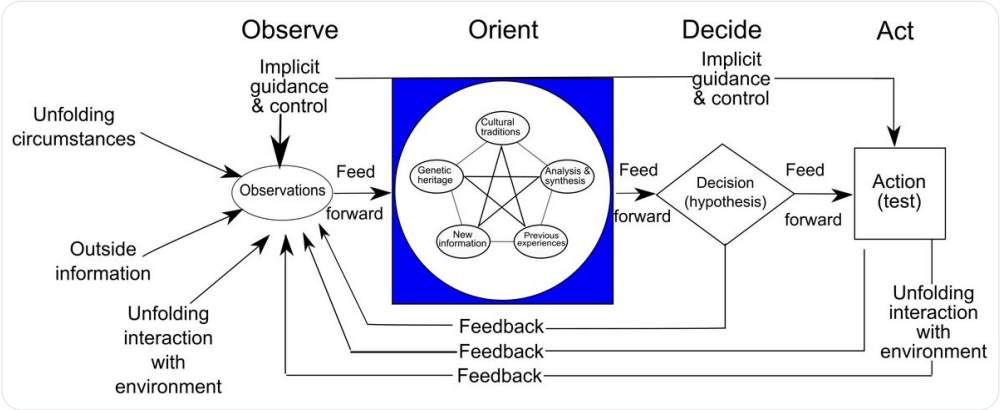
° Comments precede the moves made on the same line.
Source: Poundstone 1992: 108–116.

Not really. If the last round is decided in both players' mind, and will definitely be about mutual defection, then the game before it can be considered the "true" last game. And thus, mutual defection would be the nash equilbrium there. By induction, we should see 100% defection

But that's not what happened at all. Ironically, this experiment was reformulated as a single round for normies to demonstrate the "power" of game theory. In fact, the prisoner's dilemma demonstrated its limitation -- and the limitation of focusing on a single move and decision.

In reality, the players are ever-changing, absorb information from each other, will react to moves continuously. The game is not finite in duration and the possibility space is difficult to determine -- and is non-ergodic and ever changing.

One great model of "players" and their decision making process is John Boyd's OODA loop.
It tells a story: the more you come to understand your opponent, the more predictable their decision making becomes. The more you can get into their observables the more you can control them.



Once everything about an opponent is known, and once their options/ability to surprise you disappears, the game theoretic formulation begins to dominate. Just as in that last round.

And that's where the most important concept from warfare comes into play: The death spiral.

In these simple examples, we merely discussed a static decision set. The players were equally capable. In a real war and political struggle, you must put some capital up at stake, which you can lose.

Every decision has an immediate net loss, and a risk. Including doing nothing.

What is a death spiral, and why is it that states may stake their entire existence to avoid entering a death spiral?
To answer this we will leave the realm of the abstract and discuss the middle east, three entities approaching death spirals:

Syria
Palestine
Israel
Lebanon

/End

Kahramanmaraş
Adana
Gaziantep
Şanlıurfa
Mardin
Duhok
دهوك
Mersin
İskenderun
Mosul
الموصل
Ayia Napa
Αγ. Νάπα
Lebanon
Beirut
بيروت
Damascus
دمشق
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WEST BANK
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